fn make_canonical_noise

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of make_canonical_noise. The constructor compiles finitely many approximate-DP bounds from a PrivacyGuarantee, rather than trusting a numerical tradeoff callback.

1 Hoare Triple

Precondition

The input domain is AtomDomain<f64> without NaN, the metric is AbsoluteDistance<f64>, and sensitivity d_in is finite and nonnegative. Invalid parameters are rejected. Compilation evaluates the specification at an ordered finite epsilon grid, including an accessible approximate-DP certificate endpoint. Sampled epsilon values must be finite and nonnegative and deltas must lie in $[0, 1]$. No additional caller assertion of symmetry, convexity, or monotonicity of the sampled numerical bounds is required.

Pseudocode

```
def make_canonical_noise(input_domain, input_metric, d_in, d_out):
    assert not input_domain.nan()
    assert d_in >= 0 and d_in.is_finite()

    epsilons = sorted(set(CANONICAL_EPSILONS + d_out.certificate_endpoints()))
    samples = [(epsilon, d_out.delta(epsilon)) for epsilon in epsilons]
    # q is an outward-certified LOWER bound. Reciprocal arithmetic is exact.
    constraints = [
        (RBig(1) - RBig(delta), RBig(exp_lower(epsilon))) for epsilon, delta in samples
    ]

    def tradeoff(alpha):
        return max(max(RBig(0), a - q * alpha, (a - alpha) / q) for a, q in constraints)

    fixed_point = max(a / (RBig(1) + q) for a, q in constraints)
    assert RBig(0) < fixed_point < RBig(1) / RBig(2)

    # Compile prefix minima once. The implementation stores the final minimum
    # as exact profile slack, with a conservative log core for the remainder.
    profile = log_step_profile_with_prefix_minima(samples)
    scale = RBig(d_in)

    def function(arg):
        rv = CanonicalRV(
            shift=RBig(clamp_finite(arg)),
            scale=scale,
            tradeoff=tradeoff,
            fixed_point=fixed_point,
```

```
        )
        return PartialSample.new(rv).value()

    def privacy_map(distance):
        assert 0 <= distance <= d_in
        if distance == 0:
            return PrivacyGuarantee().with_profile(perfect_privacy_log_profile())
        return PrivacyGuarantee().with_profile(profile.clone())

    return Measurement.new(
        input_domain,
        input_metric,
        MultiDP.with_profile(profile_purity(samples, d_in)),
        function,
        privacy_map,
    )
```

Postcondition

Theorem 1.1. The constructor returns a valid measurement under `MultiDP`, with a privacy map containing only the compiled step profile, or returns an error.

2 Certified compilation

Let (ϵ_i, δ_i) be the exact sampled floating-point values, interpreted as real numbers. Directed interval exponentiation followed by a lower-directed narrowing supplies a rational q_i with $1 \le q_i \le \exp(\epsilon_i)$. Unsupported numerical ranges and provider errors are rejected, not replaced by weaker bounds. Define $a_i = 1 - \delta_i$ using exact rational arithmetic and

$$f_i(\alpha) = \max\left(0,\ a_i - q_i\alpha,\ \frac{a_i - \alpha}{q_i}\right).$$

Both slopes use the same rational q_i, with its exact reciprocal. Thus f_i is the symmetric approximate-DP tradeoff for parameters $(\log(q_i), \delta_i)$: its epigraph in the unit square is described by

$$\beta \ge 0, \qquad \beta + q_i\alpha \ge a_i, \qquad \alpha + q_i\beta \ge a_i.$$

These inequalities are invariant under exchanging α and β. Each f_i is convex, nonincreasing and lies below $1 - \alpha$.

The compiled envelope $f = \max_i f_i$ is convex and nonincreasing. Its epigraph is the intersection of the symmetric epigraphs above, so it is exactly symmetric in the generalized-inverse sense, including any zero plateau. All evaluations use rational arithmetic, not independently rounded numerical approximations to the two branches.

The fixed point of f_i is $c_i = a_i/(1 + q_i)$. Put $c = \max_i c_i$. Since every f_i is nonincreasing, $f_i(c) \le c$, and a component attaining this maximum has $f_i(c) = c$. Therefore $f(c) = c$ exactly. Compilation rejects $c = 0$ (trivial distinguishability) and $c \ge 1/2$ (perfect privacy), so the compiled curve satisfies the nontrivial canonical noise theorem's preconditions.

3 Sampling and privacy

By Awan and Vadhan 2023, Theorem 3.9, the canonical distribution for the symmetric nontrivial tradeoff f supplies f-DP noise at unit sensitivity. The PSRN inverse-CDF algorithm evaluates this exact rational envelope and its exact fixed point, scales by `d_in` and shifts by the input. The finite clamping of infinite input endpoints is nonexpansive. Final rounding to a float is postprocessing. Construction-time specification errors do not occur during sampling; random-generator failures propagate and are data-independent.

Since $f \geq f_i$, this mechanism satisfies $(\log(q_i), \delta_i)$-DP and hence (ϵ_i, δ_i)-DP at every sampled point. Let $h(\epsilon)$ be its actual privacy profile. This profile is nonincreasing, regardless of whether the numerical samples were monotone. For every $\epsilon \geq 0$,

$$h(\epsilon) \leq \min_{i:\epsilon_i \leq \epsilon} h(\epsilon_i) \leq \min_{i:\epsilon_i \leq \epsilon} \delta_i = S(\epsilon).$$

An empty minimum is defined to be one; beyond the final point the last prefix minimum is retained. Thus the step envelope is a globally valid certificate, not merely an interpolation assumed valid between calibration points.

The log-native profile stores the final minimum as exact additive slack and uses directed logarithms for the nonnegative remainder. Its reported bounds therefore enclose S conservatively; when the core is zero, the final minimum is returned exactly. This is an encoding of the step envelope, not a transfer of source delta from an unrelated input representation.

For positive distances no greater than `d_in`, the privacy map returns a fresh guarantee with only this profile. Zero distance returns a perfect privacy log profile; it does not evaluate a rounded expression $1 - \alpha$. The descriptor advertises a profile, and advertises purity only when a zero delta was sampled at a finite epsilon or the only admissible distance is zero. No input RDP, zCDP or Gaussian-DP representation is copied to the output. Consequently each successful privacy-map result certifies the sampled mechanism for all admissible distances.

References

Awan, Jordan and Salil Vadhan (2023). "Canonical Noise Distributions and Private Hypothesis Tests". In: *The Annals of Statistics* 51.2, pp. 547–572.